CUADebug: Diagnosing and Repairing Computer-Use Agent Failures

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Paper Project Code Data

1 Motivation

- Computer-use agents act on screenshots with mouse and keyboard actions.
- When a run fails, the cause is often several steps earlier.

2 CUAErrorBench

204 failed OSWorld runs from 3 agents, each labeled with its root-cause step and error type.



Reasoning errors dominate; perception and grounding cause 30%.

3 Error Taxonomy

5 families, 30 subtypes

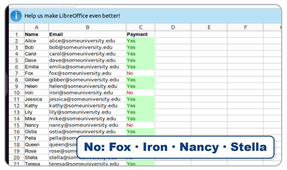
- **Perception** — 5 subtypes
  Misread what was visible?
- **Grounding** — 4 subtypes
  Right intent, wrong target?
- **Reasoning** — 13 subtypes
  Wrong plan chosen or kept?
- **External / System** — 7 subtypes
  Environment blocked progress?
- **Others** — 1 subtype
  Infeasible task

4 A Real Failure

REAL CASE **Task (abridged): Find unpaid tuition e-mails and add them to the receiver field**
Environment: LibreOffice Calc → Thunderbird

A Failure trajectory

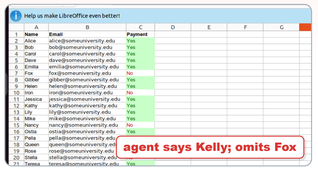
1 after t=5 Source record



The payment column identifies exactly four unpaid students; row 12 is Kathy with **Yes**.

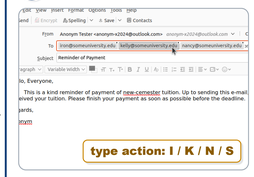
⚠ ROOT CAUSE

2 t=6 Root reasoning



The agent invents **Kelly** at row 12 and drops Fox, even though the visible row is Kathy / Yes.

3 t=10 Recipient entry



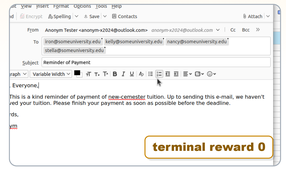
The four-address action carries the false belief into Thunderbird; the complete chips are visible at t=12.

4 1 more action



⚠ OBSERVED FAILURE

5 t=12 Final state



The draft still contains Kelly and still omits Fox when the trajectory declares completion.

The run fails at step 12, but the mistake is made at step 6, where the agent lists **Kelly** instead of **Fox**.

5 CUADebugger

1 Failed trajectory package

TASK Add unpaid students' e-mails to the receiver field
APPS Calc → Thunderbird

Input: screenshots + reasoning/action trace

- t=5 payment record
- t=6 ROOT invents Kelly; misses Fox
- t=10 wrong recipients
- Iron, Kelly, Nancy, Stella; no Fox
- t=12 FAIL declares success; reward 0

INPUT task + trajectory + terminal failure

2 Tool-guided inspection & localization

```
get_step_details(6) +
get_step_details(10)
```

t=6: before | t=10: result | t=6 Kelly / no Fox

before / after | response + action | later evidence

Candidate inspection path: inspect t=6 → compare t=5/6 → inspect t=10 → localize t=6

Root: t=6 reasoning invents Kelly and omits Fox.

OUTPUT root-step candidate + grounded evidence

3 Structured RCA record

Input: candidate step + inspected evidence

ROOT STEP	`root_error_step = 6` Invents Kelly; omits Fox
TAXONOMY	`taxonomy_tag = P1` Visual hallucination
EVIDENCE	t=5: Fox = No; t=6: Kelly/no Fox; t=10: wrong list
CORRECTION	Use all No rows: Fox, Iron, Nancy, Stella

OUTPUT structured RCA record

4 Debugger-guided re-rollout

Matched continual: Claude 4.5 Sonnet; restart t=51

Continual re-rollout: Recovery / Overall

- Direct continuation: **12.50%** / **61.22%**
- Self-Debug: **13.89%** / **61.77%**
- Machine RCA: **21.53%** / **64.82%**
- Ours + episodic memory: **25.69%** / **66.48%**

Takeaway
RCA outperforms reflection; episodic memory adds **+4.16 pp recovery (+1.66 pp overall)**.

OUTPUT re-rolled trajectory + outcome

Optional Episodic Memory (+4.16 pp recovery / +1.66 pp overall)

1 · Distill failures	2 · Build query	3 · Retrieve	4 · Verify
Trigger · failed action · correction · evidence	Task · app · state · suspected action	Top-k transferable lessons	Check screenshots and trace; discard mismatches

It inspects the before/after screenshots of selected steps, finds the earliest error, and writes a report that guides a new run.

Its report for the run above

Root step	Step 6: lists Kelly instead of Fox
Error type	P1, visual hallucination
Correction	Use the four unpaid rows: Fox, Iron, Nancy, Stella
Re-run	Fails without the report (score 0), succeeds with it (score 1)

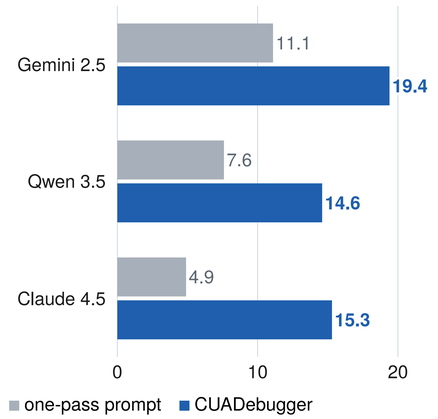
6 Results

11.1 → 19.4%
root cause found

61.77 → 68.14%
OSWorld accuracy

Root-cause analysis

type and step both correct (%)



Re-execution

accuracy on 361 OSWorld tasks (%)

	Single	Continual
Baseline	61.77	61.22
Self-debug	62.33	61.77
Machine RCA	67.59	64.82
CUADebug	**68.14**	**66.48**
Human diagnosis	*68.98*	*67.87*

Recovered failures (%)

